

Training-Free Subject-Wise Model Merging for Scalable Deployment of EEG Foundation Models

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Abstract—Recent EEG foundation models have shown promising capability in learning generalizable representations from large-scale unlabeled EEG recordings. However, substantial inter-subject variability still limits their direct deployment to downstream EEG tasks, where subject-specific adaptation is often required for reliable performance. This practice produces one adapted model per subject, leading to rapidly increasing storage and deployment costs in large-subject scenarios. To address this limitation, we investigate training-free post-hoc merging of subject-adapted EEG foundation models. Given multiple subject-specific models adapted from the same pretrained EEG foundation model, the goal is to obtain a single unified model while preserving subject-level performance. We propose a subject-wise weighted model merging strategy, which assigns different merging weights to different subject-adapted models to reduce weight-space conflicts induced by inter-subject distribution shifts. Experiments on two motor imagery EEG benchmarks, BNCI2014004 and BNCI2015001, show that the proposed method achieves average accuracies of 80.52% and 75.00%, outperforming standard weight averaging by 1.88% and 2.02%, respectively. Compared with the average cross-subject transfer performance of individual subject-adapted models, the proposed merged model further improves accuracy by 6.60% and 10.90% on the two datasets. Although fully individualized subject-specific models remain a strong empirical upper-bound solution, the proposed method substantially narrows the performance gap while requiring only one deployed model. These results demonstrate that training-free post-hoc model merging is a promising strategy for scalable and storage-efficient deployment of EEG foundation models.

Index Terms—EEG foundation model, model merging, subject adaptation, motor imagery, scalable deployment

I. INTRODUCTION

Large-scale self-supervised pretraining has recently become an effective strategy for EEG representation learning. Representative EEG foundation models, including BENDR [1], LaBraM [2], and CBraMod [3], show that generic neural representations can be learned from large collections of unlabeled EEG recordings and transferred to downstream tasks. This trend is important for brain-computer interface (BCI) research, where labeled EEG data are often scarce, noisy, and costly to collect.

Despite this progress, EEG decoding remains strongly affected by inter-subject variability. Different subjects may exhibit distinct neural patterns, electrode characteristics, and session-dependent shifts, so a pretrained EEG foundation model is often further adapted to each subject before deployment. Subject-specific adaptation is effective, but it also produces one downstream model per subject. When the same system must serve many subjects, storing and maintaining a

separate adapted checkpoint for each user becomes increasingly expensive and operationally inconvenient.

This deployment issue is not fully addressed by the current EEG foundation model literature. Existing BCI fusion strategies usually combine signals, features, or decisions. In contrast, our goal is to merge *personalized model parameters* after subject-specific adaptation. More concretely, given multiple subject-adapted models originating from the same pretrained EEG foundation model, we ask whether they can be merged into a single deployable model without additional training and while preserving as much subject-level utility as possible.

This formulation is motivated by recent progress in model merging and weight-space geometry. Prior work on model soups [4], task arithmetic [5], permutation-aware model merging [6], and linear mode connectivity [7] suggests that independently adapted neural networks can sometimes be combined in parameter space when they remain within a compatible low-loss basin. In our setting, all subject-specific models share the same pretrained initialization, which makes post-hoc merging a plausible direction. However, direct equal-weight averaging can still be suboptimal because subject-specific updates may conflict due to inter-subject distribution shift.

To address this issue, we study a simple yet effective subject-wise weighted merging strategy. Instead of assigning identical importance to all subject-adapted models, we associate each subject model with a scalar merge coefficient that modulates how strongly its adaptation is injected into the final merged model. This design keeps the method training-free at merge time and introduces only minimal additional complexity.

The main contributions of this work are as follows. First, we formulate scalable deployment of subject-adapted EEG foundation models as a training-free post-hoc model merging problem. Second, we propose a subject-wise weighted parameter merging method that explicitly accounts for heterogeneous subject contributions. Third, on BNCI2014004 and BNCI2015001, we show that the proposed merged model consistently outperforms standard averaging baselines and substantially narrows the gap to fully individualized deployment while requiring only one deployed model.

II. RELATED WORK

A. EEG Foundation Models

EEG foundation models seek to learn reusable neural representations from large-scale unlabeled or weakly labeled recordings. BENDR [1] demonstrated early that self-

supervised pretraining with Transformer-based sequence modeling can improve EEG transfer learning. More recent models such as LaBraM [2] and CBraMod [3] further scale data volume, model size, and architectural inductive bias, leading to stronger general-purpose EEG features. However, the downstream use of these models still commonly relies on subject-specific fine-tuning, especially in motor imagery tasks where inter-subject variability remains substantial.

B. Model Merging

Post-hoc parameter merging has recently emerged as a lightweight alternative to joint retraining. Model soups [4] show that averaging fine-tuned models can improve generalization without increasing inference cost. Task arithmetic [5] interprets fine-tuning updates as task vectors that can be combined in weight space. More generally, work on permutation symmetries and low-loss connectivity [6], [7] provides geometric intuition for why separately adapted models can sometimes be merged successfully. Our work is inspired by this line of research, but differs in both setting and objective: rather than merging generic task-specialized models, we merge subject-adapted EEG foundation models and focus on preserving performance across multiple subjects with a single deployment checkpoint.

III. METHOD

A. Problem Formulation

Let θ_0 denote a pretrained EEG foundation model and let θ_s denote the model obtained after adapting θ_0 to subject s , where $s \in \{1, \dots, S\}$. The objective is to construct a single merged model θ_{merge} from $\{\theta_s\}_{s=1}^S$ such that the merged model remains broadly useful across subjects while avoiding the need to deploy S separate adapted checkpoints.

For convenience, we define the subject-specific adaptation vector

$$\Delta_s = \theta_s - \theta_0. \quad (1)$$

Each Δ_s captures how subject s modifies the pretrained model.

B. Baseline Merging Strategies

The first baseline is direct equal-weight parameter averaging:

$$\theta_{\text{avg}} = \frac{1}{S} \sum_{s=1}^S \theta_s. \quad (2)$$

This strategy is simple and completely training-free, but it implicitly assumes that all subject-adapted models should contribute equally to the final solution.

The second baseline is task-vector averaging:

$$\theta_{\text{tv}} = \theta_0 + \frac{1}{S} \sum_{s=1}^S \Delta_s. \quad (3)$$

This view emphasizes the merge of adaptation directions relative to the common pretrained backbone rather than direct averaging of absolute parameters.

C. Subject-Wise Weighted Merging

Our method retains the same task-vector perspective but introduces a subject-specific coefficient λ_s for each adapted model:

$$\theta_{\text{swm}} = \theta_0 + \frac{1}{S} \sum_{s=1}^S \lambda_s \Delta_s, \quad (4)$$

where λ_s modulates the contribution of subject s to the final merged model. Setting $\lambda_s = 1$ for all s reduces (4) to uniform task-vector averaging. In this work, we use a single scalar coefficient per subject, which keeps the merged model compact and easy to interpret.

The intuition is straightforward. Subject-adapted models do not contribute equally to a common deployment model because their updates reflect different subject-specific distributions. Some adaptations are more compatible with the shared parameter basin, while others may introduce stronger interference when averaged indiscriminately. The coefficients λ_s therefore provide a lightweight mechanism to suppress conflicting updates and emphasize subject models that better align with the overall merged solution.

D. Practical Properties

The proposed method is training-free at merge time. Once subject-adapted models and their merge coefficients are available, the final checkpoint is obtained by a single pass over parameters, with no further backpropagation. This property is attractive for deployment because only one merged model needs to be stored online. In contrast, fully individualized deployment requires retaining all subject-specific models. The current formulation also remains flexible: more fine-grained coefficients, such as layer-wise or block-wise weights, can be incorporated in future work without changing the basic framework.

IV. EXPERIMENTAL SETUP

A. Datasets

We evaluate on two public motor imagery EEG benchmarks. BNCI2014004 corresponds to BCI Competition IV Dataset 2b [8] and contains data from nine subjects performing binary motor imagery classification. BNCI2015001 contains data from twelve subjects and is a widely used benchmark for cross-subject motor imagery analysis [9]. These two datasets provide complementary testbeds for studying subject variability and multi-subject deployment.

B. Evaluation Protocols

All subject-specific models are adapted from the same pretrained EEG foundation model, after which they are merged or evaluated using the following protocols.

Equal-weight model average: all subject-adapted models are averaged directly in parameter space and the resulting model is evaluated on all subjects.

Task-vector average: subject adaptation vectors are averaged relative to the common pretrained initialization and then added back to the base model.

Average cross-subject transfer of individualized models: each subject-adapted model is evaluated on all subjects, and the resulting accuracies are averaged. This quantifies how well individualized models transfer when applied beyond their own subjects.

Own-subject individualized performance: each subject-adapted model is evaluated on its corresponding subject only. We treat this setting as an empirical upper bound under fully individualized deployment.

Proposed subject-wise weighted merge: a single merged model is constructed using (4) with dataset-specific subject coefficients.

We report average classification accuracy and standard deviation. In this initial draft, the emphasis is on the merged-model behavior and deployment trade-off; the search strategy for the subject-wise coefficients can be refined and described in more detail in the next iteration.

V. RESULTS AND DISCUSSION

A. Main Results

Table I summarizes the results on both datasets. The proposed subject-wise weighted merging method achieves the best merged-model performance in both cases. On BNCI2014004, it reaches 80.52%, improving over equal-weight averaging by 1.88 percentage points and over the average cross-subject transfer performance of individualized models by 6.60 percentage points. On BNCI2015001, it reaches 75.00%, improving over equal-weight averaging by 2.02 percentage points and over average cross-subject transfer by 10.90 percentage points.

These results support two observations. First, subject-adapted models are individually strong on their own subjects but are much less reliable when transferred to other subjects. This is evident from the gap between the own-subject individualized results and the average cross-subject transfer of the same models. Second, a single merged model can recover a substantial portion of the benefit of subject-specific adaptation when the merge is made subject-aware rather than uniform.

B. Comparison to the Individualized Upper Bound

The own-subject setting remains the strongest empirical solution, because each model is evaluated on the same subject used for its adaptation. Nevertheless, the proposed method significantly narrows the deployment-performance gap while using only one checkpoint. On BNCI2014004, the gap between the proposed merged model and the individualized upper bound is only 1.79 percentage points. On BNCI2015001, the remaining gap is larger at 6.13 percentage points, indicating that the dataset exhibits stronger inter-subject diversity and is therefore harder to compress into a single shared model. Even in that more challenging case, the merged model still provides a favorable compromise between accuracy and scalability.

C. Discussion

The comparison with uniform averaging suggests that not all subject adaptations should be treated equally during merging.

Equal-weight averaging can over-emphasize conflicting subject updates, especially when the subject distributions differ substantially. Introducing subject-wise coefficients gives a simple mechanism for controlling this interference without adding training cost at merge time.

The contrast between the two datasets is also informative. On BNCI2014004, the proposed method nearly matches the individualized upper bound, which suggests that the adapted models occupy a relatively mergeable region in parameter space. On BNCI2015001, task-vector averaging drops sharply and the gap to the upper bound remains larger, which implies a more difficult merging scenario with stronger subject conflict. This pattern supports the central motivation of the paper: multi-subject EEG deployment should be analyzed not only as a transfer-learning problem, but also as a weight-space compatibility problem.

Finally, the current method intentionally stays simple. It uses a single scalar coefficient per subject and does not rely on additional retraining, permutation matching, or ensemble-time averaging. This makes the approach easy to implement and useful as a practical baseline for future EEG foundation model deployment studies.

VI. CONCLUSION

This paper studies training-free post-hoc merging of subject-adapted EEG foundation models for scalable multi-subject deployment. We formulate the problem at the parameter level and propose a subject-wise weighted merging strategy that modulates how much each subject adaptation contributes to the final unified model. Experiments on BNCI2014004 and BNCI2015001 show that the proposed method consistently outperforms standard averaging baselines and substantially reduces the gap to fully individualized deployment while requiring only one deployed checkpoint. These results suggest that model merging is a promising direction for storage-efficient and scalable EEG foundation model deployment.

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TABLE I
MAIN RESULTS ON THE TWO MOTOR IMAGERY DATASETS. VALUES ARE REPORTED AS AVERAGE ACCURACY $\pm$ STANDARD DEVIATION (%).

Dataset	Equal avg.	Task-vector	Cross-subj. indiv.	Proposed	Own-subj. UB
BNCI2014004	78.64 $\pm$ 10.52	78.64 $\pm$ 10.52	73.92 $\pm$ 3.47	80.52 $\pm$ 10.48	82.31 $\pm$ 12.66
BNCI2015001	72.98 $\pm$ 13.62	54.11 $\pm$ 4.85	64.10 $\pm$ 10.06	75.00 $\pm$ 13.15	81.13 $\pm$ 13.49

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